Công cụ - pandas

Thư viện `pandas` cung cấp các cấu trúc dữ liệu hiệu suất cao, dễ sử dụng và các công cụ phân tích dữ liệu. Cấu trúc dữ liệu chính là `DataFrame`, bạn có thể coi nó như một bảng 2D trong bộ nhớ (giống như một bảng tính, với tên cột và nhãn hàng). Nhiều tính năng có sẵn trong Excel cũng có sẵn theo cách lập trình, chẳng hạn như tạo bảng xoay (pivot tables), tính toán các cột dựa trên các cột khác, vẽ đồ thị, v.v. Bạn cũng có thể nhóm các hàng theo giá trị cột hoặc nối các bảng giống như trong SQL. Pandas cũng rất tuyệt vời trong việc xử lý chuỗi thời gian (time series).

Điều kiện tiên quyết:

- NumPy – nếu bạn chưa quen với NumPy, chúng tôi khuyên bạn nên xem qua hướng dẫn về NumPy ngay bây giờ.

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Thiết lập (Setup)

Đầu tiên, hãy import `pandas`. Mọi người thường import nó với tên viết tắt là `pd`:

```
import pandas as pd
```

Đối tượng `Series`

Thư viện `pandas` chứa các cấu trúc dữ liệu hữu ích sau:

- Đối tượng `Series`, chúng ta sẽ thảo luận ngay bây giờ. Một đối tượng `Series` là một mảng 1D, tương tự như một cột trong bảng tính (với tên cột và nhãn hàng).
- Đối tượng `DataFrame`. Đây là một bảng 2D, tương tự như một bảng tính (với tên cột và nhãn hàng).
- Đối tượng `Panel`. Bạn có thể coi `Panel` như một từ điển của các `DataFrame`. Những đối tượng này ít được sử dụng hơn, vì vậy chúng ta sẽ không thảo luận ở đây.

Tạo một `Series`

Hãy bắt đầu bằng việc tạo đối tượng `Series` đầu tiên của chúng ta!

```
s = pd.Series([2,-1,3,5])
s
```

```
0    2
1   -1
2    3
3    5
dtype: int64
```

Tương tự như một `ndarray` 1D

Các đối tượng `Series` hoạt động rất giống với các `ndarray` một chiều của NumPy, và bạn thường có thể truyền chúng dưới dạng tham số cho các hàm của NumPy:

```
import numpy as np
np.exp(s)
```

```
0      7.389056
1      0.367879
2     20.085537
3    148.413159
dtype: float64
```

Các phép toán số học trên `Series` cũng khả thi, và chúng được áp dụng *từng phần tử* (elementwise), giống như đối với các `ndarray`:

```
s + [1000,2000,3000,4000]
```

```
0    1002
1    1999
2    3003
3    4005
dtype: int64
```

Tương tự như NumPy, nếu bạn cộng một số duy nhất vào một `Series`, số đó sẽ được cộng vào tất cả các phần tử trong `Series`. Điều này được gọi là *broadcasting*:

```
s + 1000
```

```
0    1002
1     999
2    1003
3    1005
dtype: int64
```

Điều này cũng đúng với tất cả các phép toán nhị phân như `*` hoặc `/`, và thậm chí cả các phép toán điều kiện:

```
s < 0
```

```
0    False
1     True
2    False
3    False
dtype: bool
```

Nhãn chỉ mục (Index labels)

Mỗi phần tử trong một đối tượng `Series` có một định danh duy nhất được gọi là *nhãn chỉ mục* (index label). Theo mặc định, nó chỉ đơn giản là thứ hạng của phần tử trong `Series` (bắt đầu từ `0`) nhưng bạn cũng có thể thiết lập các nhãn chỉ mục theo cách thủ công:

```
s2 = pd.Series([68, 83, 112, 68], index=["alice", "bob", "charles", "darwin"])
s2
```

```
alice       68
bob         83
charles    112
darwin      68
dtype: int64
```

You can then use the `Series` just like a `dict`:

```
s2["bob"]
```

```
83
```

To make it clear when you are accessing by label or by integer location, it is recommended to always use the `loc` attribute when accessing by label, and the `iloc` attribute when accessing by integer location:

```
s2.loc["bob"]
```

```
83
```

```
s2.iloc[1]
```

```
83
```

Slicing a `Series` also slices the index labels:

```
s2.iloc[1:3]
```

```
bob         83
charles    112
dtype: int64
```

This can lead to unexpected results when using the default numeric labels, so be careful:

```
surprise = pd.Series([1000, 1001, 1002, 1003])
surprise
```

```
0    1000
1    1001
2    1002
3    1003
dtype: int64
```

```
surprise_slice = surprise[2:]
surprise_slice
```

```
2    1002
3    1003
dtype: int64
```

Oh, look! The first element has index label `2`. The element with index label `0` is absent from the slice:

```
try:
    surprise_slice[0]
except KeyError as e:
    print("Key error:", e)
```

```
Key error: 0
```

But remember that you can access elements by integer location using the `iloc` attribute. This illustrates another reason why it's always better to use `loc` and `iloc` to access `Series` objects:

```
surprise_slice.iloc[0]
```

```
1002
```

Init from `dict`

You can create a `Series` object from a `dict`. The keys will be used as index labels:

```
weights = {"alice": 68, "bob": 83, "colin": 86, "darwin": 68}
s3 = pd.Series(weights)
s3
```

```
alice     68
bob       83
colin     86
darwin    68
dtype: int64
```

You can control which elements you want to include in the `Series` and in what order by explicitly specifying the desired `index`:

```
s4 = pd.Series(weights, index = ["colin", "alice"])
s4
```

```
colin    86
alice    68
dtype: int64
```

Automatic alignment

When an operation involves multiple `Series` objects, `pandas` automatically aligns items by matching index labels.

```
print(s2.keys())
print(s3.keys())

s2 + s3
```

```
Index(['alice', 'bob', 'charles', 'darwin'], dtype='object')
Index(['alice', 'bob', 'colin', 'darwin'], dtype='object')
alice      136.0
bob        166.0
charles      NaN
colin        NaN
darwin     136.0
dtype: float64
```

The resulting `Series` contains the union of index labels from `s2` and `s3`. Since `"colin"` is missing from `s2` and `"charles"` is missing from `s3`, these items have a `NaN` result value (i.e. Not-a-Number means *missing*).

Automatic alignment is very handy when working with data that may come from various sources with varying structure and missing items. But if you forget to set the right index labels, you can have surprising results:

```
s5 = pd.Series([1000,1000,1000,1000])
print("s2 =", s2.values)
print("s5 =", s5.values)

s2 + s5
```

```
s2 = [ 68  83 112  68]
s5 = [1000 1000 1000 1000]
alice     NaN
bob       NaN
charles   NaN
darwin    NaN
0         NaN
1         NaN
2         NaN
3         NaN
dtype: float64
```

Pandas could not align the `Series`, since their labels do not match at all, hence the full `NaN` result.

Init with a scalar

You can also initialize a `Series` object using a scalar and a list of index labels: all items will be set to the scalar.

```
meaning = pd.Series(42, ["life", "universe", "everything"])
meaning
```

```
life          42
universe      42
everything    42
dtype: int64
```

`Series` name

A `Series` can have a `name`:

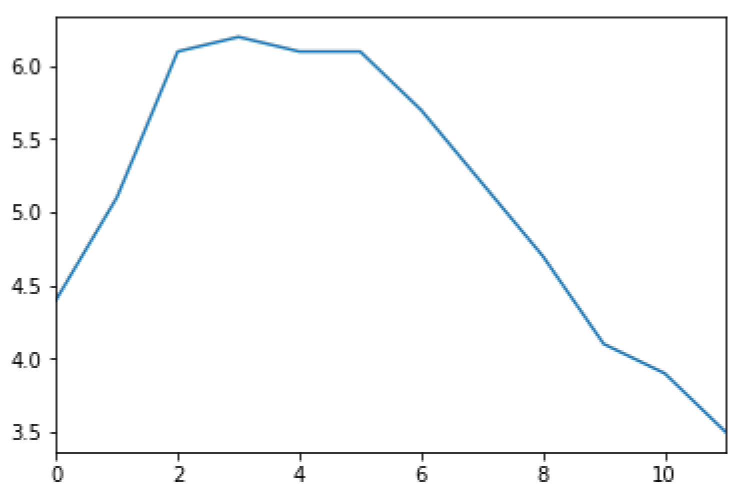
```
s6 = pd.Series([83, 68], index=["bob", "alice"], name="weights")
s6
```

```
bob      83
alice    68
Name: weights, dtype: int64
```

Plotting a `Series`

Pandas makes it easy to plot `Series` data using matplotlib (for more details on matplotlib, check out the matplotlib tutorial). Just import matplotlib and call the `plot()` method:

```
import matplotlib.pyplot as plt
temperatures = [4.4,5.1,6.1,6.2,6.1,6.1,5.7,5.2,4.7,4.1,3.9,3.5]
s7 = pd.Series(temperatures, name="Temperature")
s7.plot()
plt.show()
```



There are *many* options for plotting your data. It is not necessary to list them all here: if you need a particular type of plot (histograms, pie charts, etc.), just look for it in the excellent Visualization section of pandas' documentation, and look at the example code.

Handling time

Many datasets have timestamps, and pandas is awesome at manipulating such data:

- it can represent periods (such as 2016Q3) and frequencies (such as "monthly"),
- it can convert periods to actual timestamps, and *vice versa*,
- it can resample data and aggregate values any way you like,
- it can handle timezones.

Time range

Let's start by creating a time series using `pd.date_range()`. It returns a `DatetimeIndex` containing one datetime per hour for 12 hours starting on October 29th 2016 at 5:30pm.

```
dates = pd.date_range('2016/10/29 5:30pm', periods=12, freq='h')
dates
```

```
DatetimeIndex(['2016-10-29 17:30:00', '2016-10-29 18:30:00',
               '2016-10-29 19:30:00', '2016-10-29 20:30:00',
               '2016-10-29 21:30:00', '2016-10-29 22:30:00',
               '2016-10-29 23:30:00', '2016-10-30 00:30:00',
               '2016-10-30 01:30:00', '2016-10-30 02:30:00',
               '2016-10-30 03:30:00', '2016-10-30 04:30:00'],
              dtype='datetime64[ns]', freq='H')
```

This `DatetimeIndex` may be used as an index in a `Series`:

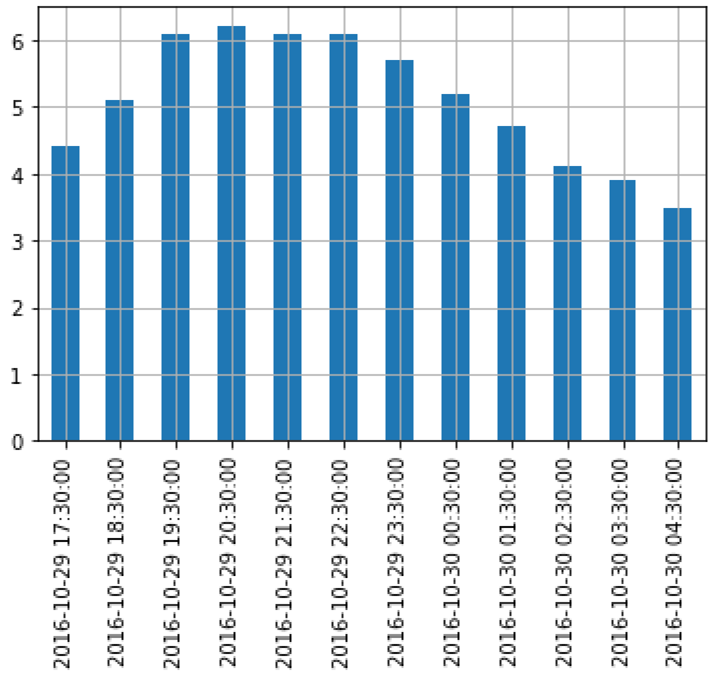
```
temp_series = pd.Series(temperatures, dates)
temp_series
```

```
2016-10-29 17:30:00    4.4
2016-10-29 18:30:00    5.1
2016-10-29 19:30:00    6.1
2016-10-29 20:30:00    6.2
2016-10-29 21:30:00    6.1
2016-10-29 22:30:00    6.1
2016-10-29 23:30:00    5.7
2016-10-30 00:30:00    5.2
2016-10-30 01:30:00    4.7
2016-10-30 02:30:00    4.1
2016-10-30 03:30:00    3.9
2016-10-30 04:30:00    3.5
Freq: H, dtype: float64
```

Let's plot this series:

```
temp_series.plot(kind="bar")

plt.grid(True)
plt.show()
```



Resampling

Pandas lets us resample a time series very simply. Just call the `resample()` method and specify a new frequency:

```
temp_series_freq_2h = temp_series.resample("2h")
temp_series_freq_2h
```

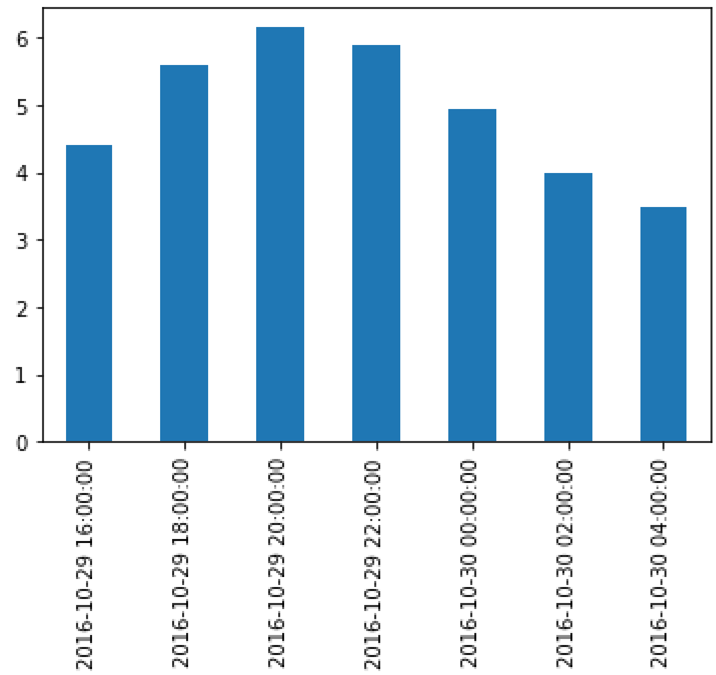
```
<pandas.core.resample.DatetimeIndexResampler object at 0x10c2c3130>
```

The resampling operation is actually a deferred operation, which is why we did not get a `Series` object, but a `DatetimeIndexResampler` object instead. To actually perform the resampling operation, we can simply call the `mean()` method. Pandas will compute the mean of every pair of consecutive hours:

```
temp_series_freq_2h = temp_series_freq_2h.mean()
```

Let's plot the result:

```
temp_series_freq_2h.plot(kind="bar")
plt.show()
```



Note how the values have automatically been aggregated into 2-hour periods. If we look at the 6-8pm period, for example, we had a value of `5.1` at 6:30pm, and `6.1` at 7:30pm. After resampling, we just have one value of `5.6`, which is the mean of `5.1` and `6.1`. Rather than computing the mean, we could have used any other aggregation function, for example we can decide to keep the minimum value of each period:

```
temp_series_freq_2h = temp_series.resample("2h").min()
temp_series_freq_2h
```

```
2016-10-29 16:00:00    4.4
2016-10-29 18:00:00    5.1
2016-10-29 20:00:00    6.1
2016-10-29 22:00:00    5.7
2016-10-30 00:00:00    4.7
2016-10-30 02:00:00    3.9
2016-10-30 04:00:00    3.5
Freq: 2H, dtype: float64
```

Or, equivalently, we could use the `apply()` method instead:

```
temp_series_freq_2h = temp_series.resample("2h").apply("min")
temp_series_freq_2h
```

```
2016-10-29 16:00:00    4.4
2016-10-29 18:00:00    5.1
2016-10-29 20:00:00    6.1
```

```
2016-10-29 22:00:00    5.7
2016-10-30 00:00:00    4.7
2016-10-30 02:00:00    3.9
2016-10-30 04:00:00    3.5
Freq: 2H, dtype: float64
```

Upsampling and interpolation

It was an example of downsampling. We can also upsample (i.e. increase the frequency), but it will create holes in our data:

```
temp_series_freq_15min = temp_series.resample("15Min").mean()
temp_series_freq_15min.head(n=10) # `head` displays the top n values
```

```
2016-10-29 17:30:00    4.4
2016-10-29 17:45:00    NaN
2016-10-29 18:00:00    NaN
2016-10-29 18:15:00    NaN
2016-10-29 18:30:00    5.1
2016-10-29 18:45:00    NaN
2016-10-29 19:00:00    NaN
2016-10-29 19:15:00    NaN
2016-10-29 19:30:00    6.1
2016-10-29 19:45:00    NaN
Freq: 15T, dtype: float64
```

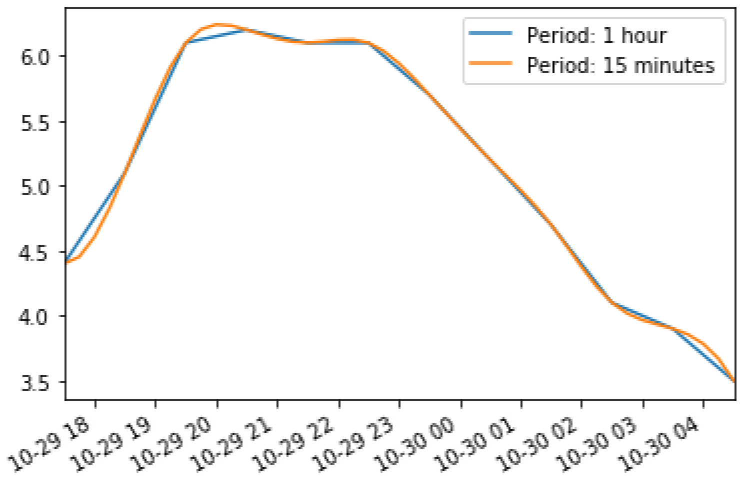
One solution is to fill the gaps by interpolating. We just call the `interpolate()` method. The default is to use linear interpolation, but we can also select another method, such as cubic interpolation:

```
temp_series_freq_15min = temp_series.resample("15Min").interpolate(method="cubi
temp_series_freq_15min.head(n=10)
```

```
2016-10-29 17:30:00    4.400000
2016-10-29 17:45:00    4.452911
2016-10-29 18:00:00    4.605113
2016-10-29 18:15:00    4.829758
2016-10-29 18:30:00    5.100000
2016-10-29 18:45:00    5.388992
2016-10-29 19:00:00    5.669887
2016-10-29 19:15:00    5.915839
2016-10-29 19:30:00    6.100000
2016-10-29 19:45:00    6.203621
Freq: 15T, dtype: float64
```

```
temp_series.plot(label="Period: 1 hour")
temp_series_freq_15min.plot(label="Period: 15 minutes")
```

```
plt.legend()
plt.show()
```



Timezones

By default, datetimes are *naive*: they are not aware of timezones, so 2016-10-30 02:30 might mean October 30th 2016 at 2:30am in Paris or in New York. We can make datetimes timezone *aware* by calling the `tz_localize()` method:

```
temp_series_ny = temp_series.tz_localize("America/New_York")
temp_series_ny
```

```
2016-10-29 17:30:00-04:00    4.4
2016-10-29 18:30:00-04:00    5.1
2016-10-29 19:30:00-04:00    6.1
2016-10-29 20:30:00-04:00    6.2
2016-10-29 21:30:00-04:00    6.1
2016-10-29 22:30:00-04:00    6.1
2016-10-29 23:30:00-04:00    5.7
2016-10-30 00:30:00-04:00    5.2
2016-10-30 01:30:00-04:00    4.7
2016-10-30 02:30:00-04:00    4.1
2016-10-30 03:30:00-04:00    3.9
2016-10-30 04:30:00-04:00    3.5
dtype: float64
```

Note that `-04:00` is now appended to all the datetimes. It means that these datetimes refer to UTC - 4 hours.

We can convert these datetimes to Paris time like this:

```
temp_series_paris = temp_series_ny.tz_convert("Europe/Paris")
temp_series_paris
```

```
2016-10-29 23:30:00+02:00    4.4
2016-10-30 00:30:00+02:00    5.1
2016-10-30 01:30:00+02:00    6.1
2016-10-30 02:30:00+02:00    6.2
2016-10-30 02:30:00+01:00    6.1
2016-10-30 03:30:00+01:00    6.1
2016-10-30 04:30:00+01:00    5.7
2016-10-30 05:30:00+01:00    5.2
2016-10-30 06:30:00+01:00    4.7
2016-10-30 07:30:00+01:00    4.1
2016-10-30 08:30:00+01:00    3.9
2016-10-30 09:30:00+01:00    3.5
dtype: float64
```

You may have noticed that the UTC offset changes from `+02:00` to `+01:00`: this is because France switches to winter time at 3am that particular night (time goes back to 2am). Notice that 2:30am occurs twice! Let's go back to a naive representation (if you log some data hourly using local time, without storing the timezone, you might get something like this):

```
temp_series_paris_naive = temp_series_paris.tz_localize(None)
temp_series_paris_naive
```

```
2016-10-29 23:30:00    4.4
2016-10-30 00:30:00    5.1
2016-10-30 01:30:00    6.1
2016-10-30 02:30:00    6.2
2016-10-30 02:30:00    6.1
2016-10-30 03:30:00    6.1
2016-10-30 04:30:00    5.7
2016-10-30 05:30:00    5.2
2016-10-30 06:30:00    4.7
2016-10-30 07:30:00    4.1
2016-10-30 08:30:00    3.9
2016-10-30 09:30:00    3.5
dtype: float64
```

Now `02:30` is really ambiguous. If we try to localize these naive datetimes to the Paris timezone, we get an error:

```
try:
    temp_series_paris_naive.tz_localize("Europe/Paris")
except Exception as e:
    print(type(e))
    print(e)
```

```
<class 'pytz.exceptions.AmbiguousTimeError'>
Cannot infer dst time from 2016-10-30 02:30:00, try using the 'ambiguous' argume
```

Fortunately, by using the `ambiguous` argument we can tell pandas to infer the right DST (Daylight Saving Time) based on the order of the ambiguous timestamps:

```
temp_series_paris_naive.tz_localize("Europe/Paris", ambiguous="infer")
```

```
2016-10-29 23:30:00+02:00    4.4
2016-10-30 00:30:00+02:00    5.1
2016-10-30 01:30:00+02:00    6.1
2016-10-30 02:30:00+02:00    6.2
2016-10-30 02:30:00+01:00    6.1
2016-10-30 03:30:00+01:00    6.1
2016-10-30 04:30:00+01:00    5.7
2016-10-30 05:30:00+01:00    5.2
2016-10-30 06:30:00+01:00    4.7
2016-10-30 07:30:00+01:00    4.1
2016-10-30 08:30:00+01:00    3.9
2016-10-30 09:30:00+01:00    3.5
dtype: float64
```

Periods

The `pd.period_range()` function returns a `PeriodIndex` instead of a `DatetimeIndex`. For example, let's get all quarters in 2016 and 2017:

```
quarters = pd.period_range('2016Q1', periods=8, freq='Q')
quarters
```

```
PeriodIndex(['2016Q1', '2016Q2', '2016Q3', '2016Q4', '2017Q1', '2017Q2',
             '2017Q3', '2017Q4'],
            dtype='period[Q-DEC]')
```

Adding a number `N` to a `PeriodIndex` shifts the periods by `N` times the `PeriodIndex`'s frequency:

```
quarters + 3
```

```
PeriodIndex(['2016Q4', '2017Q1', '2017Q2', '2017Q3', '2017Q4', '2018Q1',
             '2018Q2', '2018Q3'],
            dtype='period[Q-DEC]')
```

The `asfreq()` method lets us change the frequency of the `PeriodIndex`. All periods are lengthened or shortened accordingly. For example, let's convert all the quarterly periods to monthly periods (zooming in):

```
quarters.asfreq("M")
```

```
PeriodIndex(['2016-03', '2016-06', '2016-09', '2016-12', '2017-03', '2017-06',
             '2017-09', '2017-12'],
            dtype='period[M]')
```

By default, the `asfreq` zooms on the end of each period. We can tell it to zoom on the start of each period instead:

```
quarters.asfreq("M", how="start")
```

```
PeriodIndex(['2016-01', '2016-04', '2016-07', '2016-10', '2017-01', '2017-04',
             '2017-07', '2017-10'],
            dtype='period[M]')
```

And we can zoom out:

```
quarters.asfreq("Y")
```

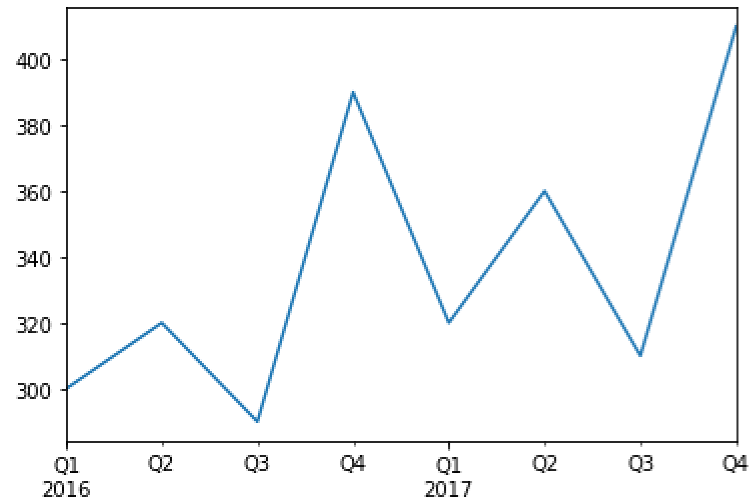
```
PeriodIndex(['2016', '2016', '2016', '2016', '2017', '2017', '2017', '2017'],
dtype='period[A-DEC]')
```

Of course, we can create a `Series` with a `PeriodIndex`:

```
quarterly_revenue = pd.Series([300, 320, 290, 390, 320, 360, 310, 410], index=c
quarterly_revenue
```

```
2016Q1    300
2016Q2    320
2016Q3    290
2016Q4    390
2017Q1    320
2017Q2    360
2017Q3    310
2017Q4    410
Freq: Q-DEC, dtype: int64
```

```
quarterly_revenue.plot(kind="line")
plt.show()
```



We can convert periods to timestamps by calling `to_timestamp`. By default, it will give us the first day of each period, but by setting `how` and `freq`, we can get the last hour of each period:

```
last_hours = quarterly_revenue.to_timestamp(how="end", freq="h")
last_hours
```

```
2016-03-31 23:59:59.999999999    300
2016-06-30 23:59:59.999999999    320
2016-09-30 23:59:59.999999999    290
2016-12-31 23:59:59.999999999    390
2017-03-31 23:59:59.999999999    320
2017-06-30 23:59:59.999999999    360
2017-09-30 23:59:59.999999999    310
2017-12-31 23:59:59.999999999    410
dtype: int64
```

And back to periods by calling `to_period`:

```
last_hours.to_period()
```

```
2016Q1    300
2016Q2    320
2016Q3    290
2016Q4    390
2017Q1    320
2017Q2    360
2017Q3    310
2017Q4    410
Freq: Q-DEC, dtype: int64
```

Pandas also provides many other time-related functions that we recommend you check out in the [documentation](). To whet your appetite, here is one way to get the last business day of each month in 2016, at 9am:

```
months_2016 = pd.period_range("2016", periods=12, freq="M")
one_day_after_last_days = months_2016.asfreq("D") + 1
last_bdays = one_day_after_last_days.to_timestamp() - pd.tseries.offsets.BDay()
last_bdays.to_period("h") + 9
```

```
PeriodIndex(['2016-01-29 09:00', '2016-02-29 09:00', '2016-03-31 09:00',
             '2016-04-29 09:00', '2016-05-31 09:00', '2016-06-30 09:00',
             '2016-07-29 09:00', '2016-08-31 09:00', '2016-09-30 09:00',
             '2016-10-31 09:00', '2016-11-30 09:00', '2016-12-30 09:00'],
            dtype='period[H]')
```

`DataFrame` objects

A DataFrame object represents a spreadsheet, with cell values, column names and row index labels. You can define expressions to compute columns based on other columns, create pivot-tables, group rows, draw graphs, etc. You can see `DataFrame`s as dictionaries of `Series`.

Creating a `DataFrame`

You can create a DataFrame by passing a dictionary of `Series` objects:

```
people_dict = {
    "weight": pd.Series([68, 83, 112], index=["alice", "bob", "charles"]),
    "birthyear": pd.Series([1984, 1985, 1992], index=["bob", "alice", "charles'
    "children": pd.Series([0, 3], index=["charles", "bob"]),
    "hobby": pd.Series(["Biking", "Dancing"], index=["alice", "bob"]),
}
people = pd.DataFrame(people_dict)
people
```

	weight	birthyear	children	hobby
alice	68	1985	NaN	Biking
bob	83	1984	3.0	Dancing
charles	112	1992	0.0	NaN

A few things to note:

- the `Series` were automatically aligned based on their index,
- missing values are represented as `NaN`,

- `Series` names are ignored (the name `"year"` was dropped),
- `DataFrame`s are displayed nicely in Jupyter notebooks, woohoo!

You can access columns pretty much as you would expect. They are returned as `Series` objects:

```
people["birthyear"]
```

```
alice      1985
bob        1984
charles    1992
Name: birthyear, dtype: int64
```

You can also get multiple columns at once:

```
people[["birthyear", "hobby"]]
```

	birthyear	hobby
alice	1985	Biking
bob	1984	Dancing
charles	1992	NaN

If you pass a list of columns and/or index row labels to the `DataFrame` constructor, it will guarantee that these columns and/or rows will exist, in that order, and no other column/row will exist. For example:

```
d2 = pd.DataFrame(
        people_dict,
        columns=["birthyear", "weight", "height"],
        index=["bob", "alice", "eugene"]
     )
d2
```

	birthyear	weight	height
bob	1984.0	83.0	NaN
alice	1985.0	68.0	NaN
eugene	NaN	NaN	NaN

Another convenient way to create a `DataFrame` is to pass all the values to the constructor as an `ndarray`, or a list of lists, and specify the column names and row index labels separately:

```
values = [
            [1985, np.nan, "Biking",   68],
            [1984, 3,      "Dancing",  83],
            [1992, 0,      np.nan,    112]
         ]
d3 = pd.DataFrame(
        values,
        columns=["birthyear", "children", "hobby", "weight"],
        index=["alice", "bob", "charles"]
     )
d3
```

	birthyear	children	hobby	weight
alice	1985	NaN	Biking	68
bob	1984	3.0	Dancing	83
charles	1992	0.0	NaN	112

To specify missing values, you can use either `np.nan` or NumPy's masked arrays:

```
masked_array = np.ma.asarray(values, dtype=object)
masked_array[(0, 2), (1, 2)] = np.ma.masked
d3 = pd.DataFrame(
        masked_array,
        columns=["birthyear", "children", "hobby", "weight"],
        index=["alice", "bob", "charles"]
     )
d3
```

	birthyear	children	hobby	weight
alice	1985	NaN	Biking	68
bob	1984	3	Dancing	83
charles	1992	0	NaN	112

Instead of an `ndarray`, you can also pass a `DataFrame` object:

```
d4 = pd.DataFrame(
         d3,
```

```
            columns=["hobby", "children"],
            index=["alice", "bob"]
        )
d4
```

	hobby	children
alice	Biking	NaN
bob	Dancing	3

It is also possible to create a `DataFrame` with a dictionary (or list) of dictionaries (or lists):

```
people = pd.DataFrame({
    "birthyear": {"alice": 1985, "bob": 1984, "charles": 1992},
    "hobby": {"alice": "Biking", "bob": "Dancing"},
    "weight": {"alice": 68, "bob": 83, "charles": 112},
    "children": {"bob": 3, "charles": 0}
})
people
```

	birthyear	hobby	weight	children
alice	1985	Biking	68	NaN
bob	1984	Dancing	83	3.0
charles	1992	NaN	112	0.0

Multi-indexing

If all columns are tuples of the same size, then they are understood as a multi-index. The same goes for row index labels. For example:

```
d5 = pd.DataFrame(
  {
    ("public", "birthyear"):
        {("Paris","alice"): 1985, ("Paris","bob"): 1984, ("London","charles"):
    ("public", "hobby"):
        {("Paris","alice"): "Biking", ("Paris","bob"): "Dancing"},
    ("private", "weight"):
        {("Paris","alice"): 68, ("Paris","bob"): 83, ("London","charles"): 112}
    ("private", "children"):
        {("Paris", "alice"): np.nan, ("Paris","bob"): 3, ("London","charles"):
  }
```

```
)
d5
```

		public		private	
		birthyear	hobby	weight	children
Paris	**alice**	1985	Biking	68	NaN
	bob	1984	Dancing	83	3.0
London	**charles**	1992	NaN	112	0.0

You can now get a `DataFrame` containing all the `"public"` columns very simply:

```
d5["public"]
```

		birthyear	hobby
Paris	**alice**	1985	Biking
	bob	1984	Dancing
London	**charles**	1992	NaN

```
d5["public", "hobby"]  # Same result as d5["public"]["hobby"]
```

```
Paris   alice        Biking
        bob         Dancing
London  charles         NaN
Name: (public, hobby), dtype: object
```

Dropping a level

Let's look at `d5` again:

```
d5
```

		public		private	
		birthyear	hobby	weight	children
Paris	**alice**	1985	Biking	68	NaN
	bob	1984	Dancing	83	3.0
London	**charles**	1992	NaN	112	0.0

There are two levels of columns, and two levels of indices. We can drop a column level by calling `droplevel()` (the same goes for indices):

```
d5.columns = d5.columns.droplevel(level = 0)
d5
```

		birthyear	hobby	weight	children
Paris	**alice**	1985	Biking	68	NaN
	bob	1984	Dancing	83	3.0
London	**charles**	1992	NaN	112	0.0

Transposing

You can swap columns and indices using the `T` attribute:

```
d6 = d5.T
d6
```

	Paris		London
	alice	bob	charles
birthyear	1985	1984	1992
hobby	Biking	Dancing	NaN
weight	68	83	112
children	NaN	3.0	0.0

Stacking and unstacking levels

Calling the `stack()` method will push the lowest column level after the lowest index:

```
d7 = d6.stack()
d7
```

		Paris	London
birthyear	**alice**	1985	NaN
	bob	1984	NaN
	charles	NaN	1992
hobby	**alice**	Biking	NaN
	bob	Dancing	NaN
weight	**alice**	68	NaN
	bob	83	NaN
	charles	NaN	112
children	**bob**	3.0	NaN
	charles	NaN	0.0

Note that many `NaN` values appeared. This makes sense because many new combinations did not exist before (e.g. there was no `bob` in `London`).

Calling `unstack()` will do the reverse, once again creating many `NaN` values.

```
d8 = d7.unstack()
d8
```

	Paris			London		
	alice	bob	charles	alice	bob	charles
birthyear	1985	1984	NaN	NaN	NaN	1992
children	NaN	3.0	NaN	NaN	NaN	0.0
hobby	Biking	Dancing	NaN	NaN	NaN	NaN
weight	68	83	NaN	NaN	NaN	112

If we call `unstack` again, we end up with a `Series` object:

```
d9 = d8.unstack()
d9
```

```
Paris   alice    birthyear       1985
                 children         NaN
                 hobby         Biking
                 weight            68
        bob      birthyear       1984
```

```
                 children        3.0
                 hobby       Dancing
                 weight           83
         charles birthyear       NaN
                 children        NaN
                 hobby           NaN
                 weight          NaN
London   alice   birthyear       NaN
                 children        NaN
                 hobby           NaN
                 weight          NaN
         bob     birthyear       NaN
                 children        NaN
                 hobby           NaN
                 weight          NaN
         charles birthyear      1992
                 children        0.0
                 hobby           NaN
                 weight          112
dtype: object
```

The `stack()` and `unstack()` methods let you select the `level` to stack/unstack. You can even stack/unstack multiple levels at once:

```
d10 = d9.unstack(level = (0,1))
d10
```

	Paris			London		
	alice	bob	charles	alice	bob	charles
birthyear	1985	1984	NaN	NaN	NaN	1992
children	NaN	3.0	NaN	NaN	NaN	0.0
hobby	Biking	Dancing	NaN	NaN	NaN	NaN
weight	68	83	NaN	NaN	NaN	112

Most methods return modified copies

As you may have noticed, the `stack()` and `unstack()` methods do not modify the object they are called on. Instead, they work on a copy and return that copy. This is true of most methods in pandas.

Accessing rows

Let's go back to the `people` `DataFrame`:

```
people
```

	birthyear	hobby	weight	children
alice	1985	Biking	68	NaN
bob	1984	Dancing	83	3.0
charles	1992	NaN	112	0.0

The `loc` attribute lets you access rows instead of columns. The result is a `Series` object in which the `DataFrame`'s column names are mapped to row index labels:

```
people.loc["charles"]
```

```
birthyear    1992
hobby         NaN
weight        112
children      0.0
Name: charles, dtype: object
```

You can also access rows by integer location using the `iloc` attribute:

```
people.iloc[2]
```

```
birthyear    1992
hobby         NaN
weight        112
children      0.0
Name: charles, dtype: object
```

You can also get a slice of rows, and this returns a `DataFrame` object:

```
people.iloc[1:3]
```

	birthyear	hobby	weight	children
bob	1984	Dancing	83	3.0
charles	1992	NaN	112	0.0

Finally, you can pass a boolean array to get the matching rows:

```
people[np.array([True, False, True])]
```

	birthyear	hobby	weight	children
alice	1985	Biking	68	NaN
charles	1992	NaN	112	0.0

This is most useful when combined with boolean expressions:

```
people[people["birthyear"] < 1990]
```

	birthyear	hobby	weight	children
alice	1985	Biking	68	NaN
bob	1984	Dancing	83	3.0

Adding and removing columns

You can generally treat `DataFrame` objects like dictionaries of `Series`, so the following works fine:

```
people
```

	birthyear	hobby	weight	children
alice	1985	Biking	68	NaN
bob	1984	Dancing	83	3.0
charles	1992	NaN	112	0.0

```
people["age"] = 2018 - people["birthyear"]  # adds a new column "age"
people["over 30"] = people["age"] > 30      # adds another column "over 30"
birthyears = people.pop("birthyear")
del people["children"]

people
```

	hobby	weight	age	over 30
alice	Biking	68	33	True
bob	Dancing	83	34	True
charles	NaN	112	26	False

```
birthyears
```

```
alice      1985
bob        1984
charles    1992
Name: birthyear, dtype: int64
```

When you add a new column, it must have the same number of rows. Missing rows are filled with NaN, and extra rows are ignored:

```
people["pets"] = pd.Series({"bob": 0, "charles": 5, "eugene": 1})  # alice is m
people
```

	hobby	weight	age	over 30	pets
alice	Biking	68	33	True	NaN
bob	Dancing	83	34	True	0.0
charles	NaN	112	26	False	5.0

When adding a new column, it is added at the end (on the right) by default. You can also insert a column anywhere else using the `insert()` method:

```
people.insert(1, "height", [172, 181, 185])
people
```

	hobby	height	weight	age	over 30	pets
alice	Biking	172	68	33	True	NaN
bob	Dancing	181	83	34	True	0.0
charles	NaN	185	112	26	False	5.0

Assigning new columns

You can also create new columns by calling the `assign()` method. Note that this returns a new `DataFrame` object, the original is not modified:

```
people.assign(
    body_mass_index = people["weight"] / (people["height"] / 100) ** 2,
    has_pets = people["pets"] > 0
)
```

	hobby	height	weight	age	over 30	pets	body_mass_index	has_pets
alice	Biking	172	68	33	True	NaN	22.985398	False
bob	Dancing	181	83	34	True	0.0	25.335002	False
charles	NaN	185	112	26	False	5.0	32.724617	True

Note that you cannot access columns created within the same assignment:

```
try:
    people.assign(
        body_mass_index = people["weight"] / (people["height"] / 100) ** 2,
        overweight = people["body_mass_index"] > 25
    )
except KeyError as e:
    print("Key error:", e)
```

```
Key error: 'body_mass_index'
```

The solution is to split this assignment in two consecutive assignments:

```
d6 = people.assign(body_mass_index = people["weight"] / (people["height"] / 10
d6.assign(overweight = d6["body_mass_index"] > 25)
```

	hobby	height	weight	age	over 30	pets	body_mass_index	overweight
alice	Biking	172	68	33	True	NaN	22.985398	False
bob	Dancing	181	83	34	True	0.0	25.335002	True
charles	NaN	185	112	26	False	5.0	32.724617	True

Having to create a temporary variable `d6` is not very convenient. You may want to just chain the assignment calls, but it does not work because the `people` object is not actually modified by the first assignment:

```
try:
    (people
         .assign(body_mass_index = people["weight"] / (people["height"] / 100)
         .assign(overweight = people["body_mass_index"] > 25)
    )
except KeyError as e:
    print("Key error:", e)
```

```
Key error: 'body_mass_index'
```

But fear not, there is a simple solution. You can pass a function to the `assign()` method (typically a `lambda` function), and this function will be called with the `DataFrame` as a parameter:

```
(people
     .assign(body_mass_index = lambda df: df["weight"] / (df["height"] / 100) *
     .assign(overweight = lambda df: df["body_mass_index"] > 25)
)
```

	hobby	height	weight	age	over 30	pets	body_mass_index	overweight
alice	Biking	172	68	33	True	NaN	22.985398	False
bob	Dancing	181	83	34	True	0.0	25.335002	True
charles	NaN	185	112	26	False	5.0	32.724617	True

Problem solved!

Evaluating an expression

A great feature supported by pandas is expression evaluation. It relies on the `numexpr` library which must be installed.

```
people.eval("weight / (height/100) ** 2 > 25")
```

```
alice      False
bob         True
charles     True
dtype: bool
```

Assignment expressions are also supported. Let's set `inplace=True` to directly modify the `DataFrame` rather than getting a modified copy:

```
people.eval("body_mass_index = weight / (height/100) ** 2", inplace=True)
people
```

	hobby	height	weight	age	over 30	pets	body_mass_index
alice	Biking	172	68	33	True	NaN	22.985398
bob	Dancing	181	83	34	True	0.0	25.335002
charles	NaN	185	112	26	False	5.0	32.724617

You can use a local or global variable in an expression by prefixing it with `'@'`:

```
overweight_threshold = 30
people.eval("overweight = body_mass_index > @overweight_threshold", inplace=Tru
people
```

	hobby	height	weight	age	over 30	pets	body_mass_index	overweight
alice	Biking	172	68	33	True	NaN	22.985398	False
bob	Dancing	181	83	34	True	0.0	25.335002	False
charles	NaN	185	112	26	False	5.0	32.724617	True

Querying a `DataFrame`

The `query()` method lets you filter a `DataFrame` based on a query expression:

```
people.query("age > 30 and pets == 0")
```

	hobby	height	weight	age	over 30	pets	body_mass_index	overweight
bob	Dancing	181	83	34	True	0.0	25.335002	False

Sorting a `DataFrame`

You can sort a `DataFrame` by calling its `sort_index` method. By default, it sorts the rows by their index label, in ascending order, but let's reverse the order:

```
people.sort_index(ascending=False)
```

	hobby	height	weight	age	over 30	pets	body_mass_index	overweight
charles	NaN	185	112	26	False	5.0	32.724617	True
bob	Dancing	181	83	34	True	0.0	25.335002	False
alice	Biking	172	68	33	True	NaN	22.985398	False

Note that `sort_index` returned a sorted *copy* of the `DataFrame`. To modify `people` directly, we can set the `inplace` argument to `True`. Also, we can sort the columns instead of the rows by setting `axis=1`:

```
people.sort_index(axis=1, inplace=True)
people
```

	age	body_mass_index	height	hobby	over 30	overweight	pets	weight
alice	33	22.985398	172	Biking	True	False	NaN	68
bob	34	25.335002	181	Dancing	True	False	0.0	83
charles	26	32.724617	185	NaN	False	True	5.0	112

To sort the `DataFrame` by the values instead of the labels, we can use `sort_values` and specify the column to sort by:

```
people.sort_values(by="age", inplace=True)
people
```

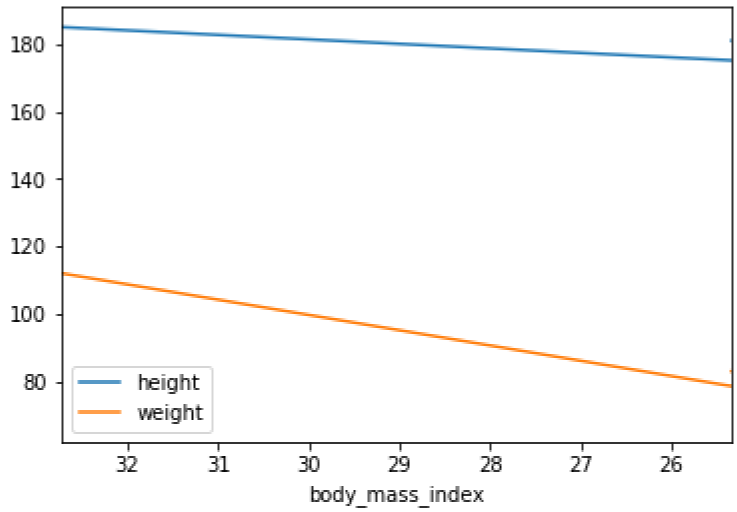
	age	body_mass_index	height	hobby	over 30	overweight	pets	weight
charles	26	32.724617	185	NaN	False	True	5.0	112
alice	33	22.985398	172	Biking	True	False	NaN	68
bob	34	25.335002	181	Dancing	True	False	0.0	83

Plotting a `DataFrame`

Just like for `Series`, pandas makes it easy to draw nice graphs based on a `DataFrame`.

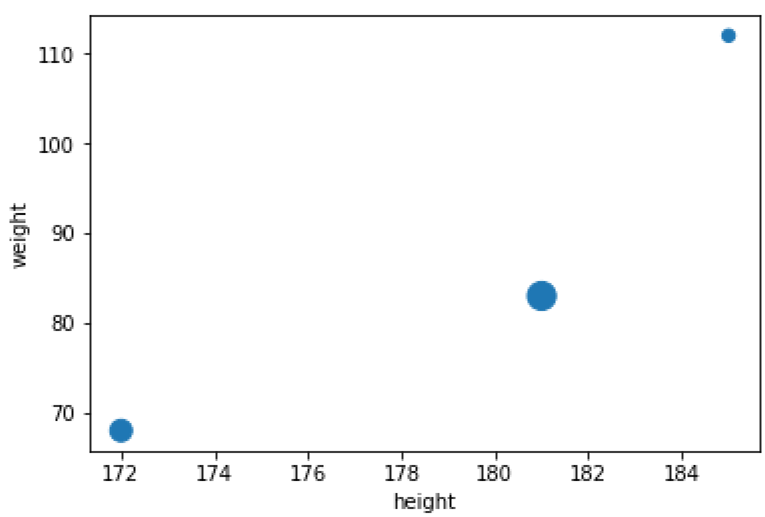
For example, it is trivial to create a line plot from a `DataFrame`'s data by calling its `plot` method:

```
people.sort_values(by="body_mass_index", inplace=True)
people.plot(kind="line", x="body_mass_index", y=["height", "weight"])
plt.show()
```



You can pass extra arguments supported by matplotlib's functions. For example, we can create scatterplot and pass it a list of sizes using the `s` argument of matplotlib's `scatter()` function:

```
people.plot(kind="scatter", x="height", y="weight", s=[40, 120, 200])
plt.show()
```



Again, there are way too many options to list here: the best option is to scroll through the Visualization page in pandas' documentation, find the plot you are interested in and look at the example code.

Operations on `DataFrame`s

Although `DataFrame`s do not try to mimic NumPy arrays, there are a few similarities. Let's create a `DataFrame` to demonstrate this:

```
grades_array = np.array([[8, 8, 9], [10, 9, 9], [4, 8, 2], [9, 10, 10]])
grades = pd.DataFrame(grades_array, columns=["sep", "oct", "nov"], index=["alic
grades
```

	sep	oct	nov
alice	8	8	9
bob	10	9	9
charles	4	8	2
darwin	9	10	10

You can apply NumPy mathematical functions on a `DataFrame`: the function is applied to all values:

```
np.sqrt(grades)
```

	sep	oct	nov
alice	2.828427	2.828427	3.000000
bob	3.162278	3.000000	3.000000
charles	2.000000	2.828427	1.414214
darwin	3.000000	3.162278	3.162278

Similarly, adding a single value to a `DataFrame` will add that value to all elements in the `DataFrame`. This is called *broadcasting*:

```
grades + 1
```

	sep	oct	nov
alice	9	9	10
bob	11	10	10
charles	5	9	3
darwin	10	11	11

Of course, the same is true for all other binary operations, including arithmetic (`*`,`/`,`**`...) and conditional (`>`, `==`...) operations:

```
grades >= 5
```

	sep	oct	nov
alice	True	True	True
bob	True	True	True
charles	False	True	False
darwin	True	True	True

Aggregation operations, such as computing the `max`, the `sum` or the `mean` of a `DataFrame`, apply to each column, and you get back a `Series` object:

```
grades.mean()
```

```
sep    7.75
oct    8.75
nov    7.50
dtype: float64
```

The `all` method is also an aggregation operation: it checks whether all values are `True` or not. Let's see during which months all students got a grade greater than `5`:

```
(grades > 5).all()
```

```
sep    False
oct     True
nov    False
dtype: bool
```

Most of these functions take an optional `axis` parameter which lets you specify along which axis of the `DataFrame` you want the operation executed. The default is `axis=0`, meaning that the operation is executed vertically (on each column). You can set `axis=1` to execute the operation horizontally (on each row). For example, let's find out which students had all grades greater than `5`:

```
(grades > 5).all(axis=1)
```

```
alice       True
bob         True
charles    False
darwin      True
dtype: bool
```

The `any` method returns `True` if any value is True. Let's see who got at least one grade 10:

```
(grades == 10).any(axis=1)
```

```
alice      False
bob         True
charles    False
darwin      True
dtype: bool
```

If you add a `Series` object to a `DataFrame` (or execute any other binary operation), pandas attempts to broadcast the operation to all *rows* in the `DataFrame`. This only works if the `Series` has the same size as the `DataFrame`s rows. For example, let's subtract the `mean` of the `DataFrame` (a `Series` object) from the `DataFrame`:

```
grades - grades.mean()  # equivalent to: grades - [7.75, 8.75, 7.50]
```

	sep	oct	nov
alice	0.25	-0.75	1.5
bob	2.25	0.25	1.5
charles	-3.75	-0.75	-5.5
darwin	1.25	1.25	2.5

We subtracted `7.75` from all September grades, `8.75` from October grades and `7.50` from November grades. It is equivalent to subtracting this `DataFrame`:

```
pd.DataFrame([[7.75, 8.75, 7.50]]*4, index=grades.index, columns=grades.columns
```

	sep	oct	nov
alice	7.75	8.75	7.5
bob	7.75	8.75	7.5
charles	7.75	8.75	7.5
darwin	7.75	8.75	7.5

If you want to subtract the global mean from every grade, here is one way to do it:

```
grades - grades.values.mean() # subtracts the global mean (8.00) from all grade
```

	sep	oct	nov
alice	0.0	0.0	1.0
bob	2.0	1.0	1.0
charles	-4.0	0.0	-6.0
darwin	1.0	2.0	2.0

Automatic alignment

Similar to `Series`, when operating on multiple `DataFrame`s, pandas automatically aligns them by row index label, but also by column names. Let's create a `DataFrame` with bonus points for each person from October to December:

```
bonus_array = np.array([[0, np.nan, 2], [np.nan, 1, 0], [0, 1, 0], [3, 3, 0]])
bonus_points = pd.DataFrame(bonus_array, columns=["oct", "nov", "dec"], index=[
bonus_points
```

	oct	nov	dec
bob	0.0	NaN	2.0
colin	NaN	1.0	0.0
darwin	0.0	1.0	0.0
charles	3.0	3.0	0.0

```
grades + bonus_points
```

	dec	nov	oct	sep
alice	NaN	NaN	NaN	NaN
bob	NaN	NaN	9.0	NaN
charles	NaN	5.0	11.0	NaN
colin	NaN	NaN	NaN	NaN
darwin	NaN	11.0	10.0	NaN

Looks like the addition worked in some cases but way too many elements are now empty. That's because when aligning the `DataFrame`s, some columns and rows were only present on one side, and thus they were considered missing on the other side (`NaN`). Then adding `NaN` to a number results in `NaN`, hence the result.

Handling missing data

Dealing with missing data is a frequent task when working with real life data. Pandas offers a few tools to handle missing data.

Let's try to fix the problem above. For example, we can decide that missing data should result in a zero, instead of `NaN`. We can replace all `NaN` values by any value using the `fillna()` method:

```
(grades + bonus_points).fillna(0)
```

	dec	nov	oct	sep
alice	0.0	0.0	0.0	0.0
bob	0.0	0.0	9.0	0.0
charles	0.0	5.0	11.0	0.0
colin	0.0	0.0	0.0	0.0
darwin	0.0	11.0	10.0	0.0

It's a bit unfair that we're setting grades to zero in September, though. Perhaps we should decide that missing grades are missing grades, but missing bonus points should be replaced by zeros:

```
fixed_bonus_points = bonus_points.fillna(0)
fixed_bonus_points.insert(0, "sep", 0)
fixed_bonus_points.loc["alice"] = 0
grades + fixed_bonus_points
```

	dec	nov	oct	sep
alice	NaN	9.0	8.0	8.0
bob	NaN	9.0	9.0	10.0
charles	NaN	5.0	11.0	4.0
colin	NaN	NaN	NaN	NaN
darwin	NaN	11.0	10.0	9.0

That's much better: although we made up some data, we have not been too unfair.

Another way to handle missing data is to interpolate. Let's look at the `bonus_points` `DataFrame` again:

```
bonus_points
```

	oct	nov	dec
bob	0.0	NaN	2.0
colin	NaN	1.0	0.0
darwin	0.0	1.0	0.0
charles	3.0	3.0	0.0

Now let's call the `interpolate` method. By default, it interpolates vertically (`axis=0`), so let's tell it to interpolate horizontally (`axis=1`).

```
bonus_points.interpolate(axis=1)
```

	oct	nov	dec
bob	0.0	1.0	2.0
colin	NaN	1.0	0.0
darwin	0.0	1.0	0.0
charles	3.0	3.0	0.0

Bob had 0 bonus points in October, and 2 in December. When we interpolate for November, we get the mean: 1 bonus point. Colin had 1 bonus point in November, but we do not know how many bonus points he had in September, so we cannot interpolate, this is why there is still a missing value in October after interpolation. To fix this, we can set the September bonus points to 0 before interpolation.

```
better_bonus_points = bonus_points.copy()
better_bonus_points.insert(0, "sep", 0)
better_bonus_points.loc["alice"] = 0
better_bonus_points = better_bonus_points.interpolate(axis=1)
better_bonus_points
```

	sep	oct	nov	dec
bob	0.0	0.0	1.0	2.0
colin	0.0	0.5	1.0	0.0
darwin	0.0	0.0	1.0	0.0
charles	0.0	3.0	3.0	0.0
alice	0.0	0.0	0.0	0.0

Great, now we have reasonable bonus points everywhere. Let's find out the final grades:

```
grades + better_bonus_points
```

	dec	nov	oct	sep
alice	NaN	9.0	8.0	8.0
bob	NaN	10.0	9.0	10.0
charles	NaN	5.0	11.0	4.0
colin	NaN	NaN	NaN	NaN
darwin	NaN	11.0	10.0	9.0

It is slightly annoying that the September column ends up on the right. This is because the `DataFrame`s we are adding do not have the exact same columns (the `grades` `DataFrame` is missing the `"dec"` column), so to make things predictable, pandas orders the final columns alphabetically. To fix this, we can simply add the missing column before adding:

```
grades["dec"] = np.nan
final_grades = grades + better_bonus_points
final_grades
```

	sep	oct	nov	dec
alice	8.0	8.0	9.0	NaN
bob	10.0	9.0	10.0	NaN
charles	4.0	11.0	5.0	NaN
colin	NaN	NaN	NaN	NaN
darwin	9.0	10.0	11.0	NaN

There's not much we can do about December and Colin: it's bad enough that we are making up bonus points, but we can't reasonably make up grades (well, I guess some teachers probably do). So let's call the `dropna()` method to get rid of rows that are full of `NaN`s:

```
final_grades_clean = final_grades.dropna(how="all")
final_grades_clean
```

	sep	oct	nov	dec
alice	8.0	8.0	9.0	NaN
bob	10.0	9.0	10.0	NaN
charles	4.0	11.0	5.0	NaN
darwin	9.0	10.0	11.0	NaN

Now let's remove columns that are full of `NaN`s by setting the `axis` argument to `1`:

```
final_grades_clean = final_grades_clean.dropna(axis=1, how="all")
final_grades_clean
```

	sep	oct	nov
alice	8.0	8.0	9.0
bob	10.0	9.0	10.0
charles	4.0	11.0	5.0
darwin	9.0	10.0	11.0

Aggregating with `groupby`

Similar to the SQL language, pandas allows grouping your data into groups to run calculations over each group.

First, let's add some extra data about each person so we can group them, and let's go back to the `final_grades` `DataFrame` so we can see how `NaN` values are handled:

```
final_grades["hobby"] = ["Biking", "Dancing", np.nan, "Dancing", "Biking"]
final_grades
```

	sep	oct	nov	dec	hobby
alice	8.0	8.0	9.0	NaN	Biking
bob	10.0	9.0	10.0	NaN	Dancing
charles	4.0	11.0	5.0	NaN	NaN
colin	NaN	NaN	NaN	NaN	Dancing
darwin	9.0	10.0	11.0	NaN	Biking

Now let's group data in this `DataFrame` by hobby:

```
grouped_grades = final_grades.groupby("hobby")
grouped_grades
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x1a4f80250>
```

We are ready to compute the average grade per hobby:

```
grouped_grades.mean()
```

	sep	oct	nov	dec
hobby				
Biking	8.5	9.0	10.0	NaN
Dancing	10.0	9.0	10.0	NaN

That was easy! Note that the `NaN` values have simply been skipped when computing the means.

Pivot tables

Pandas supports spreadsheet-like pivot tables that allow quick data summarization. To illustrate this, let's create a simple `DataFrame`:

```
bonus_points
```

	oct	nov	dec
bob	0.0	NaN	2.0
colin	NaN	1.0	0.0
darwin	0.0	1.0	0.0
charles	3.0	3.0	0.0

```
more_grades = final_grades_clean.stack().reset_index()
more_grades.columns = ["name", "month", "grade"]
more_grades["bonus"] = [np.nan, np.nan, np.nan, 0, np.nan, 2, 3, 3, 0, 0, 1, 0]
more_grades
```

	name	month	grade	bonus
0	alice	sep	8.0	NaN
1	alice	oct	8.0	NaN
2	alice	nov	9.0	NaN
3	bob	sep	10.0	0.0
4	bob	oct	9.0	NaN
5	bob	nov	10.0	2.0
6	charles	sep	4.0	3.0
7	charles	oct	11.0	3.0
8	charles	nov	5.0	0.0
9	darwin	sep	9.0	0.0
10	darwin	oct	10.0	1.0
11	darwin	nov	11.0	0.0

Now we can call the `pd.pivot_table()` function for this `DataFrame`, asking to group by the `name` column. By default, `pivot_table()` computes the mean of each numeric column:

```
pd.pivot_table(more_grades[["name", "grade", "bonus"]], index="name")
```

	bonus	grade
name		
alice	NaN	8.333333
bob	1.000000	9.666667
charles	2.000000	6.666667
darwin	0.333333	10.000000

We can change the aggregation function by setting the `aggfunc` argument, and we can also specify the list of columns whose values will be aggregated:

```
pd.pivot_table(more_grades, index="name", values=["grade", "bonus"], aggfunc="m
```

	bonus	grade
name		
alice	NaN	9.0
bob	2.0	10.0
charles	3.0	11.0
darwin	1.0	11.0

We can also specify the `columns` to aggregate over horizontally, and request the grand totals for each row and column by setting `margins=True`:

```
pd.pivot_table(more_grades, index="name", values="grade", columns="month", marg
```

month	nov	oct	sep	All
name				
alice	9.00	8.0	8.00	8.333333
bob	10.00	9.0	10.00	9.666667
charles	5.00	11.0	4.00	6.666667
darwin	11.00	10.0	9.00	10.000000
All	8.75	9.5	7.75	8.666667

Finally, we can specify multiple index or column names, and pandas will create multi-level indices:

```
pd.pivot_table(more_grades, index=("name", "month"), margins=True)
```

		bonus	grade
name	**month**		
alice	**nov**	NaN	9.00
	oct	NaN	8.00
	sep	NaN	8.00
bob	**nov**	2.000	10.00
	oct	NaN	9.00
	sep	0.000	10.00
charles	**nov**	0.000	5.00
	oct	3.000	11.00
	sep	3.000	4.00
darwin	**nov**	0.000	11.00
	oct	1.000	10.00
	sep	0.000	9.00
All		1.125	8.75

Overview functions

When dealing with large `DataFrames`, it is useful to get a quick overview of its content. Pandas offers a few functions for this. First, let's create a large `DataFrame` with a mix of numeric values, missing values and text values. Notice how Jupyter displays only the corners of the `DataFrame`:

```
much_data = np.fromfunction(lambda x,y: (x+y*y)%17*11, (10000, 26))
large_df = pd.DataFrame(much_data, columns=list("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
large_df[large_df % 16 == 0] = np.nan
large_df.insert(3, "some_text", "Blabla")
large_df
```

	A	B	C	some_text	D	E	F	G	H	I	...	Q
0	NaN	11.0	44.0	Blabla	99.0	NaN	88.0	22.0	165.0	143.0	...	11.0
1	11.0	22.0	55.0	Blabla	110.0	NaN	99.0	33.0	NaN	154.0	...	22.0
2	22.0	33.0	66.0	Blabla	121.0	11.0	110.0	44.0	NaN	165.0	...	33.0
3	33.0	44.0	77.0	Blabla	132.0	22.0	121.0	55.0	11.0	NaN	...	44.0
4	44.0	55.0	88.0	Blabla	143.0	33.0	132.0	66.0	22.0	NaN	...	55.0
...	...	...	...	...	...	...	...	...	...	...	...	...
9995	NaN	NaN	33.0	Blabla	88.0	165.0	77.0	11.0	154.0	132.0	...	NaN
9996	NaN	11.0	44.0	Blabla	99.0	NaN	88.0	22.0	165.0	143.0	...	11.0
9997	11.0	22.0	55.0	Blabla	110.0	NaN	99.0	33.0	NaN	154.0	...	22.0
9998	22.0	33.0	66.0	Blabla	121.0	11.0	110.0	44.0	NaN	165.0	...	33.0
9999	33.0	44.0	77.0	Blabla	132.0	22.0	121.0	55.0	11.0	NaN	...	44.0

10000 rows × 27 columns

The `head()` method returns the top 5 rows:

```
large_df.head()
```

	A	B	C	some_text	D	E	F	G	H	I	...	Q	
0	NaN	11.0	44.0	Blabla	99.0	NaN	88.0	22.0	165.0	143.0	...	11.0	Na
1	11.0	22.0	55.0	Blabla	110.0	NaN	99.0	33.0	NaN	154.0	...	22.0	11.
2	22.0	33.0	66.0	Blabla	121.0	11.0	110.0	44.0	NaN	165.0	...	33.0	22.
3	33.0	44.0	77.0	Blabla	132.0	22.0	121.0	55.0	11.0	NaN	...	44.0	33.
4	44.0	55.0	88.0	Blabla	143.0	33.0	132.0	66.0	22.0	NaN	...	55.0	44.

5 rows × 27 columns

Of course, there's also a `tail()` function to view the bottom 5 rows. You can pass the number of rows you want:

```
large_df.tail(n=2)
```

	A	B	C	some_text	D	E	F	G	H	I	...	Q	
9998	22.0	33.0	66.0	Blabla	121.0	11.0	110.0	44.0	NaN	165.0	...	33.0	2
9999	33.0	44.0	77.0	Blabla	132.0	22.0	121.0	55.0	11.0	NaN	...	44.0	3

2 rows × 27 columns

The `info()` method prints out a summary of each column's contents:

```
large_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 27 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   A          8823 non-null   float64
 1   B          8824 non-null   float64
 2   C          8824 non-null   float64
 3   some_text  10000 non-null  object
 4   D          8824 non-null   float64
 5   E          8822 non-null   float64
 6   F          8824 non-null   float64
 7   G          8824 non-null   float64
 8   H          8822 non-null   float64
 9   I          8823 non-null   float64
 10  J          8823 non-null   float64
 11  K          8822 non-null   float64
 12  L          8824 non-null   float64
 13  M          8824 non-null   float64
 14  N          8822 non-null   float64
```

```
 15   O           8824 non-null   float64
 16   P           8824 non-null   float64
 17   Q           8824 non-null   float64
 18   R           8823 non-null   float64
 19   S           8824 non-null   float64
 20   T           8824 non-null   float64
 21   U           8824 non-null   float64
 22   V           8822 non-null   float64
 23   W           8824 non-null   float64
 24   X           8824 non-null   float64
 25   Y           8822 non-null   float64
 26   Z           8823 non-null   float64
dtypes: float64(26), object(1)
memory usage: 2.1+ MB
```

Finally, the `describe()` method gives a nice overview of the main aggregated values over each column:

- `count`: number of non-null (not NaN) values
- `mean`: mean of non-null values
- `std`: standard deviation of non-null values
- `min`: minimum of non-null values
- `25%`, `50%`, `75%`: 25th, 50th and 75th percentile of non-null values
- `max`: maximum of non-null values

```
large_df.describe()
```

	A	B	C	D	E	
count	8823.000000	8824.000000	8824.000000	8824.000000	8822.000000	8824.00000
mean	87.977559	87.972575	87.987534	88.012466	87.983791	88.00748
std	47.535911	47.535523	47.521679	47.521679	47.535001	47.51937
min	11.000000	11.000000	11.000000	11.000000	11.000000	11.00000
25%	44.000000	44.000000	44.000000	44.000000	44.000000	44.00000
50%	88.000000	88.000000	88.000000	88.000000	88.000000	88.00000
75%	132.000000	132.000000	132.000000	132.000000	132.000000	132.00000
max	165.000000	165.000000	165.000000	165.000000	165.000000	165.00000

8 rows × 26 columns

Saving & loading

Pandas can save `DataFrame`s to various backends, including file formats such as CSV, Excel, JSON, HTML and HDF5, or to a SQL database. Let's create a `DataFrame` to demonstrate this:

```
my_df = pd.DataFrame(
    [["Biking", 68.5, 1985, np.nan], ["Dancing", 83.1, 1984, 3]],
    columns=["hobby", "weight", "birthyear", "children"],
    index=["alice", "bob"]
)
my_df
```

	hobby	weight	birthyear	children
alice	Biking	68.5	1985	NaN
bob	Dancing	83.1	1984	3.0

Saving

Let's save it to CSV, HTML and JSON:

```
my_df.to_csv("my_df.csv")
my_df.to_html("my_df.html")
my_df.to_json("my_df.json")
```

Done! Let's take a peek at what was saved:

```
from pathlib import Path

for filename in ("my_df.csv", "my_df.html", "my_df.json"):
    print("#", filename)
    print(Path(filename).read_text())
    print()
```

```
# my_df.csv
,hobby,weight,birthyear,children
alice,Biking,68.5,1985,
bob,Dancing,83.1,1984,3.0


# my_df.html
<table border="1" class="dataframe">
  <thead>
    <tr style="text-align: right;">
      <th></th>
      <th>hobby</th>
```

```
      <th>weight</th>
      <th>birthyear</th>
      <th>children</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th>alice</th>
      <td>Biking</td>
      <td>68.5</td>
      <td>1985</td>
      <td>NaN</td>
    </tr>
    <tr>
      <th>bob</th>
      <td>Dancing</td>
      <td>83.1</td>
      <td>1984</td>
      <td>3.0</td>
    </tr>
  </tbody>
</table>

# my_df.json
{"hobby":{"alice":"Biking","bob":"Dancing"},"weight":{"alice":68.5,"bob":83.1},"
```

Note that the index is saved as the first column in a CSV file, as `<th>` tags in HTML and as keys in JSON. The index column has no name by default, but you can fix that by setting `my_df.index.name` to any name you want.

Saving to other formats works very similarly, but some formats require extra libraries to be installed. For example, saving to Excel requires the openpyxl library:

```
try:
    my_df.to_excel("my_df.xlsx", sheet_name='People')
except ImportError as e:
    print(e)
```

Loading

Now let's load our CSV file back into a `DataFrame`:

```
my_df_loaded = pd.read_csv("my_df.csv", index_col=0)
my_df_loaded
```

	hobby	weight	birthyear	children
alice	Biking	68.5	1985	NaN
bob	Dancing	83.1	1984	3.0

As you might guess, there are similar `read_json`, `read_html`, `read_excel` functions as well. We can also read data straight from the Internet. For example, let's load the top 1,000 U.S. cities from GitHub:

```
us_cities = None
try:
    csv_url = "https://raw.githubusercontent.com/plotly/datasets/master/us-citi
    us_cities = pd.read_csv(csv_url, index_col=0)
    us_cities = us_cities.head()
except IOError as e:
    print(e)
us_cities
```

	State	Population	lat	lon
City				
Marysville	Washington	63269	48.051764	-122.177082
Perris	California	72326	33.782519	-117.228648
Cleveland	Ohio	390113	41.499320	-81.694361
Worcester	Massachusetts	182544	42.262593	-71.802293
Columbia	South Carolina	133358	34.000710	-81.034814

There are more options available, in particular regarding datetime format. Check out the documentation for more details.

Combining `DataFrame`s

SQL-like joins

One powerful feature of pandas is its ability to perform SQL-like joins on `DataFrame`s. Various types of joins are supported: inner joins, left/right outer joins and full joins. To illustrate this, let's start by creating a couple of simple `DataFrame`s:

```
city_loc = pd.DataFrame(
    [
        ["CA", "San Francisco", 37.781334, -122.416728],
        ["NY", "New York", 40.705649, -74.008344],
        ["FL", "Miami", 25.791100, -80.320733],
        ["OH", "Cleveland", 41.473508, -81.739791],
        ["UT", "Salt Lake City", 40.755851, -111.896657]
    ], columns=["state", "city", "lat", "lng"])
city_loc
```

	state	city	lat	lng
0	CA	San Francisco	37.781334	-122.416728
1	NY	New York	40.705649	-74.008344
2	FL	Miami	25.791100	-80.320733
3	OH	Cleveland	41.473508	-81.739791
4	UT	Salt Lake City	40.755851	-111.896657

```
city_pop = pd.DataFrame(
    [
        [808976, "San Francisco", "California"],
        [8363710, "New York", "New-York"],
        [413201, "Miami", "Florida"],
        [2242193, "Houston", "Texas"]
    ], index=[3,4,5,6], columns=["population", "city", "state"])
city_pop
```

	population	city	state
3	808976	San Francisco	California
4	8363710	New York	New-York
5	413201	Miami	Florida
6	2242193	Houston	Texas

Now let's join these `DataFrame`s using the `merge()` function:

```
pd.merge(left=city_loc, right=city_pop, on="city")
```

	state_x	city	lat	lng	population	state_y
0	CA	San Francisco	37.781334	-122.416728	808976	California
1	NY	New York	40.705649	-74.008344	8363710	New-York
2	FL	Miami	25.791100	-80.320733	413201	Florida

Note that both `DataFrame`s have a column named `state`, so in the result they got renamed to `state_x` and `state_y`.

Also, note that Cleveland, Salt Lake City and Houston were dropped because they don't exist in *both* `DataFrame`s. This is the equivalent of a SQL `INNER JOIN`. If you want a `FULL OUTER JOIN`, where no city gets dropped and `NaN` values are added, you must specify `how="outer"`:

```
all_cities = pd.merge(left=city_loc, right=city_pop, on="city", how="outer")
all_cities
```

	state_x	city	lat	lng	population	state_y
0	CA	San Francisco	37.781334	-122.416728	808976.0	California
1	NY	New York	40.705649	-74.008344	8363710.0	New-York
2	FL	Miami	25.791100	-80.320733	413201.0	Florida
3	OH	Cleveland	41.473508	-81.739791	NaN	NaN
4	UT	Salt Lake City	40.755851	-111.896657	NaN	NaN
5	NaN	Houston	NaN	NaN	2242193.0	Texas

Of course, `LEFT OUTER JOIN` is also available by setting `how="left"`: only the cities present in the left `DataFrame` end up in the result. Similarly, with `how="right"` only cities in the right `DataFrame` appear in the result. For example:

```
pd.merge(left=city_loc, right=city_pop, on="city", how="right")
```

	state_x	city	lat	lng	population	state_y
0	CA	San Francisco	37.781334	-122.416728	808976	California
1	NY	New York	40.705649	-74.008344	8363710	New-York
2	FL	Miami	25.791100	-80.320733	413201	Florida
3	NaN	Houston	NaN	NaN	2242193	Texas

If the key to join on is actually in one (or both) `DataFrame`'s index, you must use `left_index=True` and/or `right_index=True`. If the key column names differ, you must use `left_on` and `right_on`. For example:

```
city_pop2 = city_pop.copy()
city_pop2.columns = ["population", "name", "state"]
pd.merge(left=city_loc, right=city_pop2, left_on="city", right_on="name")
```

	state_x	city	lat	lng	population	name	state_y
0	CA	San Francisco	37.781334	-122.416728	808976	San Francisco	California
1	NY	New York	40.705649	-74.008344	8363710	New York	New-York

Concatenation

Rather than joining `DataFrame`s, we may just want to concatenate them. That's what `concat()` is for:

```
result_concat = pd.concat([city_loc, city_pop])
result_concat
```

	state	city	lat	lng	population
0	CA	San Francisco	37.781334	-122.416728	NaN
1	NY	New York	40.705649	-74.008344	NaN
2	FL	Miami	25.791100	-80.320733	NaN
3	OH	Cleveland	41.473508	-81.739791	NaN
4	UT	Salt Lake City	40.755851	-111.896657	NaN
3	California	San Francisco	NaN	NaN	808976.0
4	New-York	New York	NaN	NaN	8363710.0
5	Florida	Miami	NaN	NaN	413201.0
6	Texas	Houston	NaN	NaN	2242193.0

Note that this operation aligned the data horizontally (by columns) but not vertically (by rows). In this example, we end up with multiple rows having the same index (e.g. 3). Pandas handles this rather gracefully:

```
result_concat.loc[3]
```

	state	city	lat	lng	population
3	OH	Cleveland	41.473508	-81.739791	NaN
3	California	San Francisco	NaN	NaN	808976.0

Or you can tell pandas to just ignore the index:

```
pd.concat([city_loc, city_pop], ignore_index=True)
```

	state	city	lat	lng	population
0	CA	San Francisco	37.781334	-122.416728	NaN
1	NY	New York	40.705649	-74.008344	NaN
2	FL	Miami	25.791100	-80.320733	NaN
3	OH	Cleveland	41.473508	-81.739791	NaN
4	UT	Salt Lake City	40.755851	-111.896657	NaN
5	California	San Francisco	NaN	NaN	808976.0
6	New-York	New York	NaN	NaN	8363710.0
7	Florida	Miami	NaN	NaN	413201.0
8	Texas	Houston	NaN	NaN	2242193.0

Notice that when a column does not exist in a `DataFrame`, it acts as if it was filled with `NaN` values. If we set `join="inner"`, then only columns that exist in *both* `DataFrame`s are returned:

```
pd.concat([city_loc, city_pop], join="inner")
```

	state	city
0	CA	San Francisco
1	NY	New York
2	FL	Miami
3	OH	Cleveland
4	UT	Salt Lake City
3	California	San Francisco
4	New-York	New York

You can concatenate `DataFrame`s horizontally instead of vertically by setting `axis=1`:

```
pd.concat([city_loc, city_pop], axis=1)
```

	state	city	lat	lng	population	city	state
0	CA	San Francisco	37.781334	-122.416728	NaN	NaN	NaN
1	NY	New York	40.705649	-74.008344	NaN	NaN	NaN
2	FL	Miami	25.791100	-80.320733	NaN	NaN	NaN
3	OH	Cleveland	41.473508	-81.739791	808976.0	San Francisco	California
4	UT	Salt Lake City	40.755851	-111.896657	8363710.0	New York	New-York
5	NaN	NaN	NaN	NaN	413201.0	Miami	Florida

In this case it really does not make much sense because the indices do not align well (e.g. Cleveland and San Francisco end up on the same row, because they shared the index label `3`). So let's reindex the `DataFrame`s by city name before concatenating:

```
pd.concat([city_loc.set_index("city"), city_pop.set_index("city")], axis=1)
```

	state	lat	lng	population	state
city					
San Francisco	CA	37.781334	-122.416728	808976.0	California
New York	NY	40.705649	-74.008344	8363710.0	New-York
Miami	FL	25.791100	-80.320733	413201.0	Florida
Cleveland	OH	41.473508	-81.739791	NaN	NaN